Notation/Terminology:

"Random Sample"

$$X_1, X_2, \ldots, X_n$$

- variables before they are sampled, observed, and "locked in"

- they are assumed to be independent and identically distributed (iid)

random sample $=$ iid

More Notation:

Suppose that $X_1, X_2, \ldots, X_n$ is a random sample from the gamma distribution with parameters α and β.

We write

$$X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \Gamma(\alpha, \beta)$$

More Notation:

θ will denote a generic parameter

For example, $\theta = \mu$ $\theta = p$

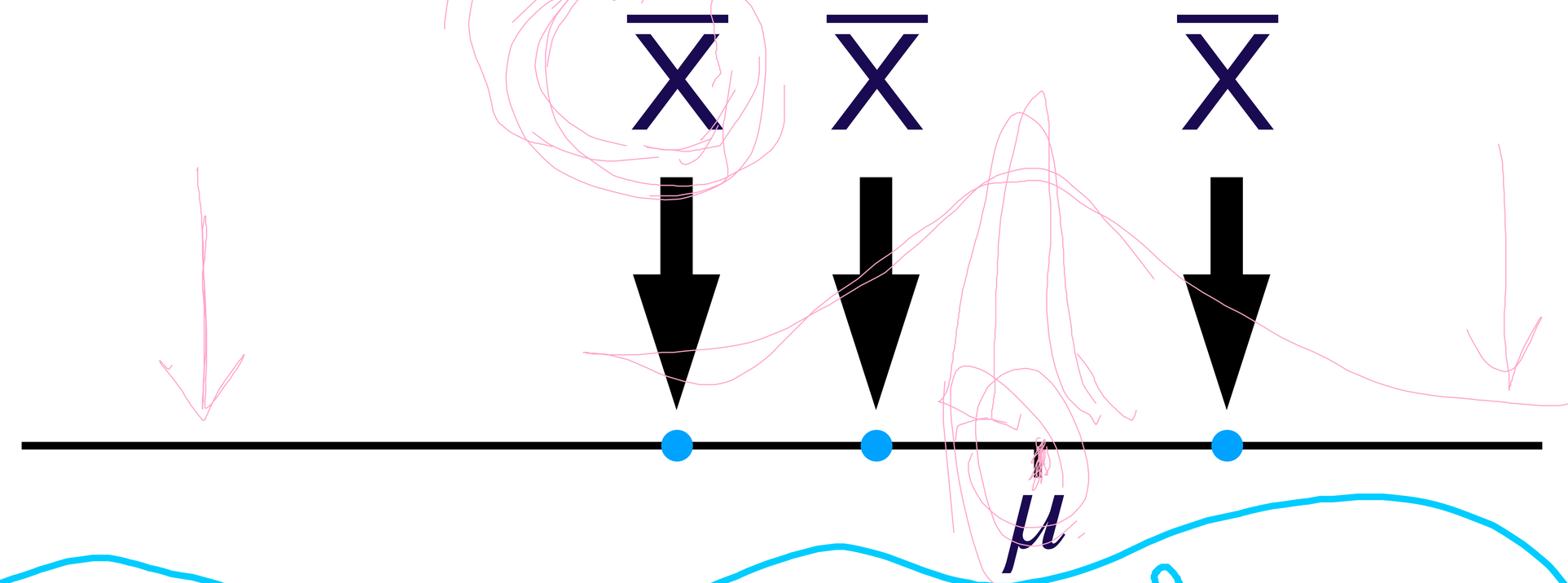
$\theta = \lambda$ $\theta = (\alpha, \beta)$

- Estimator: $\hat{\theta}$ = a random variable

  Example: $\hat{\theta} = \bar{X}$

- Estimate: $\hat{\theta}$ = an observation/number

  Example: $\hat{\theta} = \bar{x} = 42.8$



- We want our estimator of μ to be correct "on average.
- $\overline{X}$ is a random variable with its own distribution and its own mean or expected value.

We would like $E[\overline{X}] = \mu$.

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If this is true, we say that $\overline{X}$ is an unbiased estimator of μ.

In general, $\hat{\theta}$ is an unbiased estimator of θ if:

$$E[\hat{\theta}] = \theta$$

Let $X_1, X_2, \ldots, X_n$ be a random sample from any distribution with mean μ.

That is, $E[X_i] = \mu$ for $i = 1, 2, \ldots, n$.

Then

$$E[\overline{X}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]$$

$$= \frac{1}{n}\sum_{i=1}^{n} E[X_i]$$

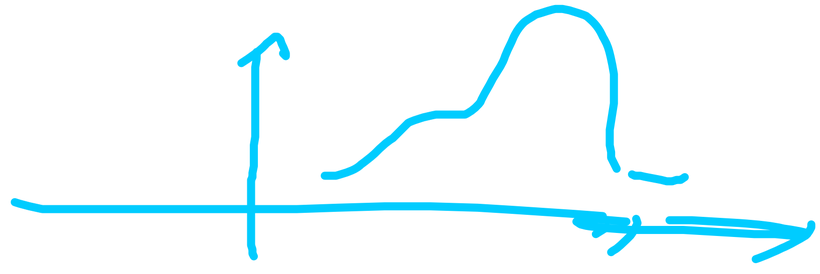
$$= \frac{1}{n}\sum_{i=1}^{n} \mu = \frac{1}{n}(\mu + \mu + \cdots + \mu) = \frac{1}{n} n\mu = \mu$$

$E[aX + bY] = aE[X] + bE[Y]$

We have shown that, no matter what distribution we are working with, if the mean is μ, $\overline{X}$ is an unbiased estimator for μ.

Example:

Suppose that

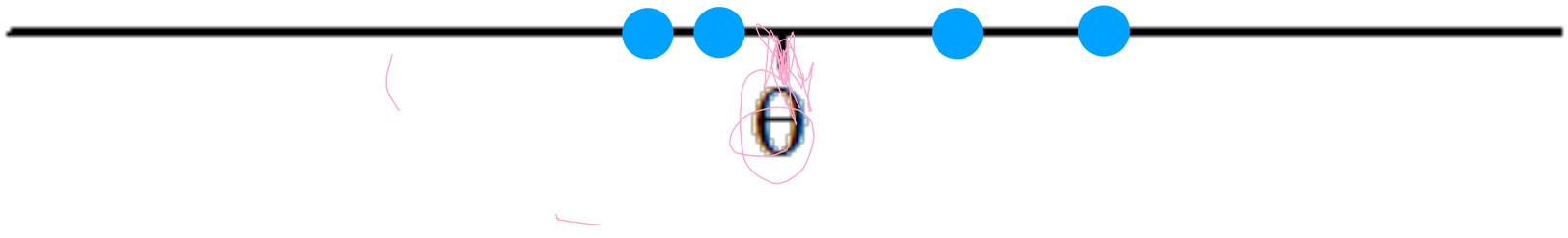


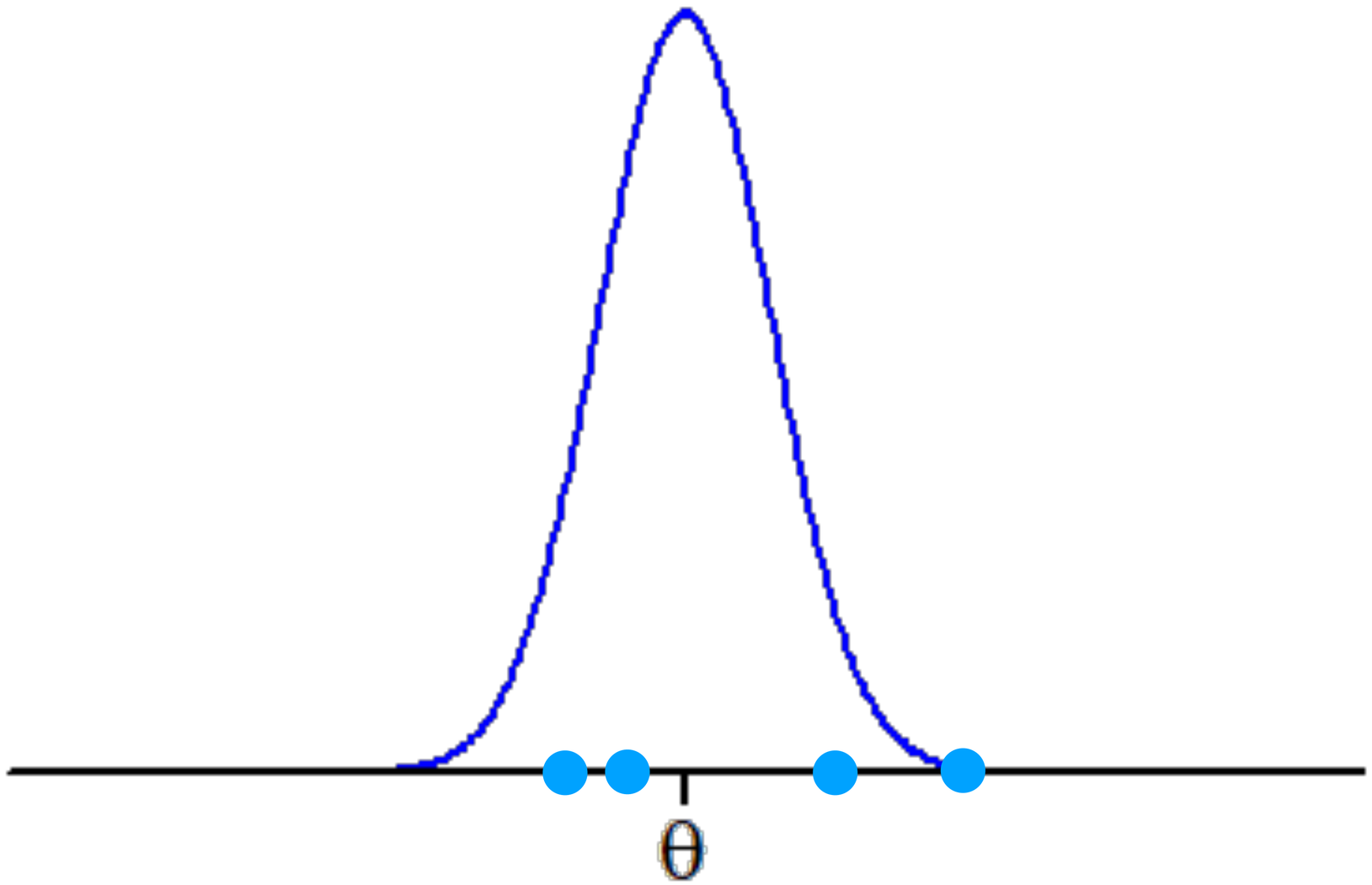
$$X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \exp(\text{rate} = \lambda)$$

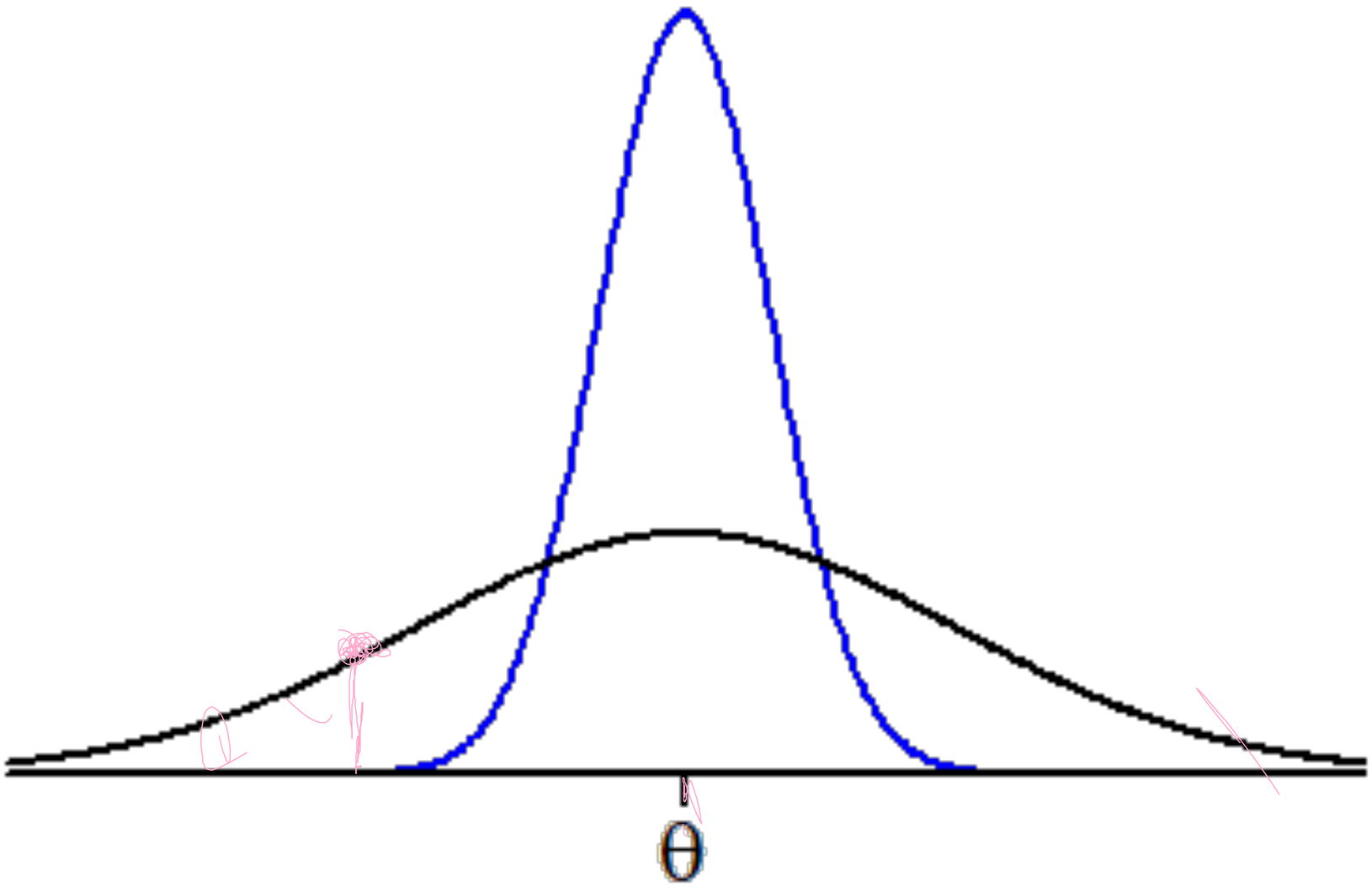
Let $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ be the sample mean.

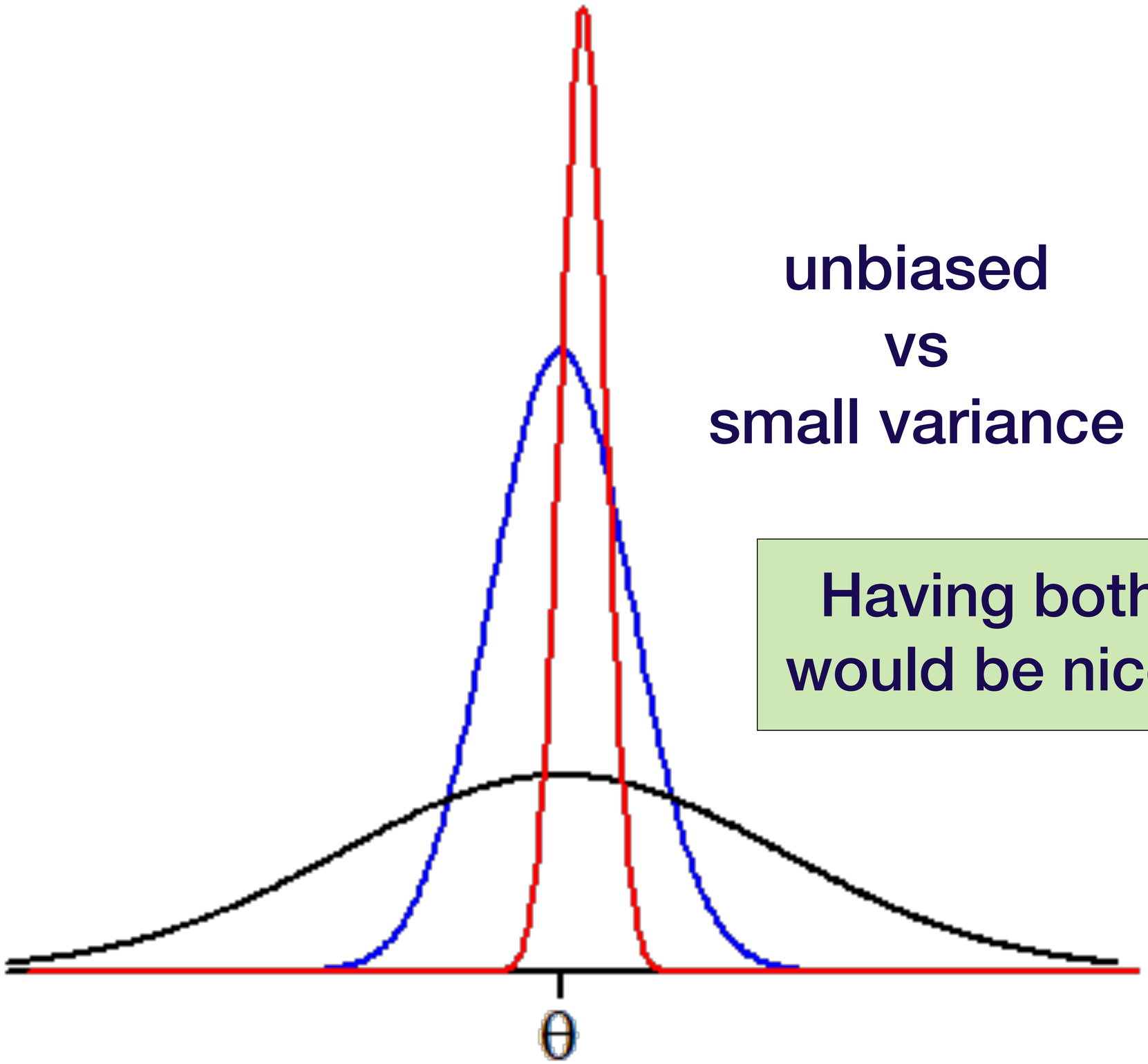
We know, for the exponential distribution, that $E[X_i] = 1/\lambda$.

Then $E[\overline{X}] = 1/\lambda$.


6


θ


θ



unbiased
vs
small variance

Having both would be nice!

Let $X_1, X_2, \ldots, X_n$ be a random sample from any distribution with mean μ and variance σ^2.

- We already know that $\overline{X}$ is an unbiased estimator for μ.
- What can we say about the variance of $\overline{X}$?

$$\text{Var}\left[\overline{X}\right] = \text{Var}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]$$

$$= \frac{1}{n^2}\text{Var}\left[\sum_{i=1}^{n} X_i\right] = \frac{1}{n^2}\sum_{i=1}^{n}\text{Var}\left[X_i\right]$$

by independence

$$\mathrm{Var}[\bar{X}] = \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}\left[X_i\right]$$

$$= \frac{1}{n^2} \sum_{i=1}^{n} \sigma^2$$

$$= \frac{1}{n^2} n\,\sigma^2$$

$$= \frac{\sigma^2}{n}$$

$n \uparrow \Rightarrow \mathrm{Var} \downarrow$